

Jonathan Mifundu Nzengi, Ph.D.
Control Systems, Robotics, and Automation Specialist

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PROFILE

Ph.D. in Mechanical Engineering (Université de Sherbrooke, 2025), with expertise in advanced control systems, robotics, and industrial automation. Experience in the design and implementation of adaptive and multivariable control algorithms, embedded programming (C++, Arduino, STM32, ROS), and HIL simulation in MATLAB/Simulink. Additional experience in service robotics (R&D internships at 3IT: kinesthetic teaching, artificial skin for robots), industrial automation with PLCs (Siemens TIA Portal), and advanced instrumentation. Demonstrated ability to cover the full development cycle of a mechatronic system, from modeling to experimental validation.

SKILLS

Control Systems	Classical control (PID), multivariable control (MIMO), robust control, adaptive control, active noise and vibration control, HIL (hardware-in-the-loop) simulation.
Robotics & Mechatronics	ROS, robot system design and integration, kinesthetic teaching, actuators and sensors for robotics, prototyping and functional validation.
Programming	C++, Python, Arduino, STM32CubeMX & Keil, MATLAB & Simulink, LADDER, PLC & HMI (Siemens PLCSIM, TIA Portal).
Modeling & Simulation	State-space modeling, finite elements (FEM/FEA), structural dynamics, numerical simulation.
Mechanical Design	Static and fatigue sizing of machine elements, SolidWorks, Autodesk Inventor, CAD modeling.
Instrumentation	Multichannel data acquisition, signal conditioning.
Languages	French: advanced (working language); English: professional (scientific reading).

EXPERIENCE

2021 – 2025	Doctoral Researcher — Active Control and Adaptive Systems <i>CRASH-UdeS, Université de Sherbrooke, Sherbrooke, Quebec</i> Development and validation of adaptive control algorithms (FxLMS, FxNLMS, IMC) in real time on an instrumented multichannel experimental test bench. State-space modeling and experimental identification of transfer functions of a complex physical system (double-wall acoustic partition). Implementation of MIMO control strategies; frequency-domain analysis of closed-loop performance.
2021	R&D Internship — Kinesthetic Teaching in Robotics (4 months) <i>3IT, Université de Sherbrooke, Sherbrooke, Quebec</i> Development of a demonstration-based programming interface for a collaborative robot (Baxter).
2021	R&D Internship — Artificial Skin for Robotics (4 months) <i>3IT, Université de Sherbrooke, Sherbrooke, Quebec</i> Design and prototyping of a flexible sensory covering for a robot.

2024	Technical Support — INOLIVENT Project <i>Université de Sherbrooke, Sherbrooke, Quebec</i> Support for the programming and experimental validation of a liquid ventilator system.
Winter 2026	Course Lecturer — Sensors and Instrumentation <i>Cégep de Victoriaville, Victoriaville, Quebec</i> Course design and delivery: instrumentation, signal conditioning, calibration, and data acquisition (45 h).
Winter 2024 & 2025	Course Lecturer — Mechanical Design Project <i>Cégep de Lévis, Lévis, Quebec</i> Supervision of capstone projects in mechanical design, CAD modeling, and component sizing (120 h/session).
2021 – 2023	Teaching Assistant (4 semesters) <i>Faculty of Engineering, Université de Sherbrooke, Sherbrooke, Quebec</i> Instructional support and supervision in numerical computation and simulation.
2019	Industrial Maintenance Internship (2 months) <i>REGIDESO (Water Treatment and Distribution), Kinshasa, DRC</i>

EDUCATION

2021 – 2025	Ph.D. in Mechanical Engineering <i>Université de Sherbrooke, Sherbrooke, Quebec</i> Thesis: <i>Broadband Active Control with Error Signal Compensation Applied to the Acoustic Isolation of Double-Wall Partitions.</i>
2021	Graduate Diploma of Specialized Studies (DESS) in Engineering <i>Université de Sherbrooke, Sherbrooke, Quebec</i>
2018 – 2021	Engineering Degree in Electromechanics <i>École Supérieure Privée d'Ingénierie et de Technologies, Tunis, Tunisia</i>
2014 – 2017	Preparatory Studies and Undergraduate Mechanical Engineering <i>Polytechnic Faculty, University of Kinshasa, Kinshasa, DRC</i>

PUBLICATIONS

1. Grandjean, P., Mifundu Nzengi, J., Micheau, P., & Berry, A. *Active Double Glazing With In-Cavity Compensated Microphones. Mechanical Systems and Signal Processing*, vol. 252, May 2026, p. 114246.
2. Mifundu Nzengi, J., Grandjean, P., Micheau, P., & Berry, A. (2024). *Preliminary study on active control of double-glazing with pressure compensation using an in-cavity microphone. INTER-NOISE and NOISE-CON Congress and Conference Proceedings*, Vol. 270, No. 2, pp. 9702–9711.

PROFESSIONAL CERTIFICATIONS

SolidWorks (CSWP)	Mechanical Design, Simulation, Advanced Weldments, Advanced Surfacing.
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